

Improved Modeling for Substitution Boxes with Negative Samples and Beyond

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Abstract. It is a common practice for symmetric-key ciphers is to encode a cryptanalysis problem as an instance of the Mixed Integer Linear Programming (MILP) and then run the instance with an efficient solver. For this purpose, it is essential to model the components in a way that is compatible with the MILP formulation while preserving the characteristics of the cipher. In this work, we look at the problem of efficiently encoding a substitution box (SBox for short). More specifically, we take the evaluation tables, namely, the Difference Distribution Table (DDT), the Linear Approximation Table (LAT), the Division Property Table (DPT) and the Boomerang Connectivity Table (BCT). In the current stage, the only model proposed/used in the literature is to look for the positive samples (i.e., non-zero) entries in the evaluation table and encode for that (this ultimately leads to the minimum number of active SBoxes for the target cipher). In our first contribution, we experiment with the complementary concept of using the negative samples (i.e., treating the zero entries as non-zero entries and vice versa), finally a proper recovery procedure is performed to get back to the original problem.

In our second contribution, we observe that the top row and column of each table (which are always taken into consideration by the past researchers) are actually redundant, as those are inherently taken care of through additional constraints at another level. We thus propose a simplified model that removes those row and column.

We show the efficacy of our proposals by furnishing results on the evaluation tables on multiple SBoxes. Our proposals often reduce the number of constraints for certain SBox evaluation tables (depending on the properties of the table) from the state-of-the-art results.

Keywords: Block Cipher · Substitution Box · Substitution Box Evaluation Table · Mixed Integer Linear Programming

1 Introduction

Over the past few years, multiple tools to find the security bounds for ciphers have emerged. These tools take the problem description of the classical attacks, such as the differential or linear attacks, as a constraint satisfaction or objective optimization problem. Not only does it take away the tedious manual effort, but it is also capable of fine-tuning the bounds (thereby finding a more optimal bound). Naturally, there has been a great deal of interest in this research area, as evidenced by the recent publications. Generally, two types of solvers are used: Satisfiability (SAT) solvers like Cryptominisat and Mixed Integer Linear Programming (MILP) solvers like Gurobi or CPLEX are used, though Constraint Programming (CP), Quadratically Constrained Linear Programming, or Simple Theorem Prover (STP) [25] are occasionally employed as the tool of choice.

In fact, it is highly important to evaluate the security of classical attacks. This is typically done by the cipher designers (in the paper proposing a new cipher) and independent researchers interested in that cipher.

While automated solvers undeniably reduce the workload for the cipher designers and analyzers alike, there is the problem of formatting the problem (related to the bounds of the cipher) to a form compatible with the solver. When designing new ciphers, it is standard practice to prioritize resistance against classical

cryptanalysis. One crucial aspect of assessing a cipher’s security against classical attacks is determining the minimum number of active SBoxes (Section 2.1) within the cipher⁴. Consequently, calculating the smallest number of active SBoxes for block ciphers [27, 36] has garnered significant attention in cryptographic research.

1.1 Overview of Prior Works

In this part, we briefly describe the major works in this field. The basic problem is to take an SBox evaluation table (*Difference Distribution Table* (DDT), *Linear Approximation Table* (LAT), *Division Property Table* (DPT), and *Boomerang Connectivity Table* (BCT)) and convert the information in the said table to a system of linear constraints (so that the constraints can be used in an MILP problem formulation). On top of that, a heuristic idea is to minimize the number of constraints [1, 2, 17, 24, 28, 33] (it is believed that the lower number of constraints will lead to faster runtime of the overall MILP instance), and this poses an extra challenge. Therefore, the fundamental challenges in this context are:

1. Construct an appropriate set of linear constraints that describes the given SBox evaluation table.
2. Identify a minimal subset of the said constraints that still preserves the SBox evaluation table.

To address these challenges, researchers have proposed various techniques. Sun et al. [33, 34] propose a method to convert from the SBox evaluation tables to linear constraints. Sun et al. [34] introduce a greedy method to eliminate redundant inequalities. In 2017, Sasaki and Todo [30] introduce an innovative algorithm designed to automatically reduce the size of descriptive models. In 2020, Boura and Coggia [18] develop a technique for generating new inequalities based on existing ones. Their experimental results showcase the ability to reduce inequalities compared to previous approaches. In 2017, Abdelkhalek et al. [1] present the first MILP model tailored for 8-bit SBoxes. They focus on the product-of-sum representation and employ the Espresso algorithm to generate the associated inequalities.

1.2 Contribution

Despite recent progress, it is observed that further optimization of results from previous works remains possible. In this paper, two novel approaches are presented to further optimize the state-of-the-art methods. In summary, improved models for SBoxes are proposed that are suitable for MILP-aided cryptanalysis to find the minimum number of active SBoxes. The approaches are general and can be applied to any SBox evaluation table including DDT, LAT, DPT and BCT. This paper restricts the analysis to bijective SBoxes for simplicity, though further generalization for non-bijective SBoxes is possible.

The key contributions can be summarized as:

- **Switching to negative samples.** A reformulation of the SBox constraint system is proposed to model the absence of specific transitions or correlations (i.e., the negative samples in the SBox evaluation tables). In this approach, the complement model (i.e., the maximum number of inactive SBoxes) is found, from which conversion to the actual problem (i.e., the minimum number of active SBoxes) is performed. If the solution from the MILP solver, say I , represents the maximum number of SBoxes that can be inactive, then to determine the minimal number of active SBoxes, this value I is subtracted from the total number of SBoxes in the cipher.
- **Exclusion of 0th row/column.** Through a survey of literature, it is observed that all existing works include the redundant top row and column (that correspond to the zero-to-zero transitions) in the SBox evaluation tables. However, this can be modeled through other means (the constraints which encode whether or not the SBox is active) automatically detect such conditions.

1.3 Organization

- In Section 2 the necessary prerequisites are provided, including definitions of the SBox and the cryptographic property tables used in the analysis, such as the *Difference Distribution Table* (DDT), *Linear Approximation Table* (LAT), *Division Property Table* (DPT), and *Boomerang Connectivity Table* (BCT).

⁴This method does not work on ciphers with linear structures, such as **DEFAULT** [5, Chapter 8] or **BAKSHEESH** [6].

- Section 3 reviews related work in the field, discussing previous methods for generating and reducing inequalities for SBox modeling.
- In Section 4 the first proposed approach is introduced, the concept of complementary variable encoding by focusing on modeling negative transitions (impossible propagation) to simplify the MILP model.
- The second proposed approach is presented in Section 5, which involves excluding redundant zero-to-zero transitions (the first row and column) in the SBox evaluation tables to further optimize the modeling process.
- Section 6 presents our experimental results, where we compare our proposed methods to traditional approaches by looking at the number of inequalities they produce across different SBoxes and property tables.
- Section 7 concludes the paper by summarizing our findings and discussing their implications for the automated security evaluation of ciphers.
- Additionally, Appendix A provides some relevant examples. These include description of our proposed methods applied to specific SBoxes and property tables. Indices to the SBoxes used/analyzed in our experiments are provided in Appendix B.

2 Prerequisites

2.1 Substitution Box (SBox)

A substitution box, (SBox for short), denoted S , is a mapping represented by $S : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. Any operation on an SBox can be depicted as a transformation from an input x to an output y , where both x and y belong to the finite field $\mathbb{F}_2^n$.

Consider the $2n$ -bit vector $(x_0, x_1, \dots, x_{n-1}, y_0, y_1, \dots, y_{n-1})$ over $\mathbb{F}_2^{2n}$ where x indicates the input characteristic to the SBox and y indicates the output characteristic from the SBox. Depending on the property of interest, the input-output characteristics can be differential propagation (in the context of differential attack) or linear masking (in the context of linear attack) and so on.

2.2 Difference Distribution Table (DDT)

The differential behavior of an SBox is usually characterized by its *Difference Distribution Table* (DDT). For an n -bit input and an m -bit output SBox $S : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, the DDT is a $2^n \times 2^m$ matrix whose entry at position $(\Delta x, \Delta y)$ is defined as $\text{DDT}[\Delta x, \Delta y] = \#\{x \in \mathbb{F}_2^n \mid S(x) \oplus S(x \oplus \Delta x) = \Delta y\}$. Here, $\Delta x \in \mathbb{F}_2^n$ denotes an input difference and $\Delta y \in \mathbb{F}_2^m$ an output difference, while $\oplus$ represents bitwise XOR. Each row of the DDT corresponds to a fixed input difference and sums to 2^n , reflecting that every input x contributes exactly one output difference. The first row (for $\Delta x = 0$) is trivial, with $\text{DDT}[0, 0] = 2^n$ and all other entries equal to zero. The maximum value among the nontrivial entries, called the *differential uniformity* of the SBox, determines its resistance against differential cryptanalysis [12, 14, 23].

2.3 Linear Approximation Table (LAT)

A *Linear Approximation Table* (LAT) is constructed to measure the biases of linear approximations between the input and output bits of a cryptographic function, such as an SBox. For an n -input, m -output function $F : \{0, 1\}^n \rightarrow \{0, 1\}^m$, we are interested in linear combinations of its inputs and outputs. Let $a \in \{0, 1\}^n$ be a binary vector representing a linear combination of input bits and $b \in \{0, 1\}^m$ is also a binary vector representing a linear combination of output bits. The linear approximation is expressed as:

$$a \cdot x \oplus b \cdot F(x) = 0, \text{ where: } a \cdot x = \bigoplus_{i=1}^n a_i x_i \text{ (XOR of the selected input bits), } b \cdot F(x) = \bigoplus_{j=1}^m b_j F(x_j) \text{ (XOR}$$

of the selected output bits). Now, the bias measures how often this approximation holds true compared to random guessing. We define: $P(a, b) = \Pr(a \cdot x \oplus b \cdot F(x) = 0)$, where x is chosen uniformly at random from $\{0, 1\}^n$. The bias $\epsilon(a, b)$ is, $\epsilon(a, b) = P(a, b) - \frac{1}{2}$.

The entry $\text{LAT}[a, b]$ in the table is computed as, $\text{LAT}[a, b] = N \cdot \epsilon(a, b)$, where $N = 2^n$ is the total number of possible inputs. Alternatively, $\text{LAT}[a, b] = \#\{x \in \{0, 1\}^n : a \cdot x \oplus b \cdot F(x) = 0\} - \frac{N}{2}$.

2.4 Division Property Table (DPT)

Let d represent the algebraic degree of the SBox. It is possible to select 2^{d+1} inputs to the SBox such that the XOR sum of the corresponding SBox outputs equals 0.

Let $\pi_u : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ be a bit-product function defined for any $u \in \mathbb{F}_2^n$ and $x \in \mathbb{F}_2^n$, where the i -th bits of u and x are denoted as u_i and x_i , respectively. The function $\pi_u(x)$ is defined as: $\pi_u(x) := \prod_{i=1}^n x_i^{u_i}$.

Let X be a multiset whose elements are values in $\mathbb{F}_2^n$, and k be an integer between 0 and n . If the multiset X satisfies the division property D_n^k , the following two conditions hold, the parity of $\pi_u(x)$ is always even for all $x \in X$ and for any u with a Hamming weight less than k . Also, the parity of $\pi_u(x)$ becomes indeterminate for any u with a Hamming weight greater than or equal to k . One approach is to examine the presence of monomials in the algebraic normal forms (ANFs) of $x \mapsto \pi_v(S(x))$ for $v \in \mathbb{F}_2^n$. This is represented by the set $P_S(u)$, defined as $\bigcup_{w \in \text{Succ}(u)} Q_S(w)$, where $Q_S(w) = \{v \in \mathbb{F}_2^n : \pi_v(S(x)) \text{ contains } \pi_w(x)\}$, and $\text{Succ}(u) = \{x \in \mathbb{F}_2^n : u \preceq x\}$, which forms an affine subspace of dimension $n - \text{wt}(u)$, with $\text{wt}(u)$ being the Hamming weight of u . The table representation of $P_S(u)$ for all u provides valuable insight into the resistance of the function against division-property-based attacks.

2.5 Boomerang Connectivity Table (BCT)

The *Boomerang Connectivity Table* (BCT) is a cryptanalysis tool used to evaluate the probability of successfully connecting two differential characteristics in a boomerang-style attack, particularly when the middle part of the cipher consists of a single SBox layer. In a boomerang attack, the cipher is often considered a composition of sub-ciphers. The attack's success depends on how the differential characteristics of these parts interact at their boundary. The BCT provides a systematic way to measure this interaction.

It analyzes the formation of a boomerang quartet, a set of four values (x_1, x_2, x_3, x_4) at the input of the SBox layer. These values must satisfy two conditions simultaneously:

1. The input difference from the first part of the cipher is Δ_i , such that $x_1 \oplus x_2 = \Delta_i$, and we check if $x_3 \oplus x_4 = \Delta_i$.
2. The output difference required by the second part of the cipher is ∇_o , such that $S(x_1) \oplus S(x_3) = \nabla_o$ and $S(x_2) \oplus S(x_4) = \nabla_o$.

The BCT calculates the probability of these conditions holding true for all possible pairs of (Δ_i, ∇_o) .

For an n -bit invertible SBox, the BCT is a $2^n \times 2^n$ table. Each entry in the table, for a given input difference Δ_i and output difference ∇_o , stores a count. It represents the number of initial values x that will result in a valid boomerang quartet.

Each entry of $\mathcal{T}_{BCT}(\Delta_i, \nabla_o)$, is calculated by the following formula, which counts the number of successful inputs: $\mathcal{T}_{BCT}(\Delta_i, \nabla_o) = \#\{x \in \{0, 1\}^n | S^{-1}(S(x) \oplus \nabla_o) \oplus S^{-1}(S(x \oplus \Delta_i) \oplus \nabla_o) = \Delta_i\}$. To find an entry, one must iterate through every possible input x (from 0 to $2^n - 1$) and check if the condition in the formula is met. The final value is the total count. The probability of forming a right quartet is this count divided by 2^n .

2.6 Classification of Samples

Initially, to construct a model for an SBox, we adopt a binary classification perspective of its evaluation tables. In this viewpoint, each entry in the SBox evaluation table is treated as a data point that is assigned to one of two classes (though, one may note that the DPT is already a binary table). Entries corresponding to positive values are categorized into positive classes, indicating feasible transitions contributing to potential differential or linear characteristics. Conversely, entries that take the value zero are placed into the negative class, since they represent impossible transitions that must be excluded from any valid characteristic. This classification provides a clean separation between feasible and infeasible behaviors of the SBox, enabling systematic design of linear constraints that capture its propagation properties within a MILP framework.

Example 1. The two encodings are illustrated using the DDT of the chosen 3-bit SBox S , 05276134. The DDT (which is given in Table 1(a)) is first converted into a binary table as shown in Table 1(b). In this representation, ‘+’ denotes positive samples and ‘−’ denotes negative samples.

□

Table 1: ABC SBox (conversion from DDT to binary DDT)

(a) DDT								
$\Delta x \backslash \Delta y$	0	1	2	3	4	5	6	7
0	8	0	0	0	0	0	0	0
1	0	0	0	0	0	4	0	4
2	0	0	4	0	0	4	0	0
3	0	0	4	0	0	0	0	4
4	0	2	0	2	2	0	2	0
5	0	2	0	2	2	0	2	0
6	0	2	0	2	2	0	2	0
7	0	2	0	2	2	0	2	0

(b) Binary DDT								
$\Delta x \backslash \Delta y$	0	1	2	3	4	5	6	7
0	+	-	-	-	-	-	-	-
1	-	-	-	+	-	-	+	-
2	-	-	+	-	-	+	-	-
3	-	-	+	-	-	-	-	+
4	-	+	-	+	+	-	+	-
5	-	+	-	+	+	-	+	-
6	-	+	-	+	+	-	+	-
7	-	+	-	+	+	-	+	-

2.7 Hierarchy of Structures

The modeling of an SBox is classified into a hierarchy of structures based on the depth of integration of its evaluation tables. This approach allows systematic construction of MILP models that capture the cryptographic properties of the SBox at varying levels of detail and complexity. Within the context, only the *primary* and *secondary* structures are discussed. However, further hierarchical levels are needed to formulate for the entire MILP model that encodes for multiple rounds of a cipher where each round consists of multiple SBoxes in parallel (*tertiary*) and a makes use of MILP solver friendly optimization in the final hierarchy (*quaternary*), both of which are kept out of scope. Refer to Appendix A for examples of primary and secondary structures.

Primary Structure In the primary structure, the focus is on directly modeling the propagation properties of an SBox using its classical cryptographic evaluation tables; such as the DDT, (absolute) LAT, DPT and BCT (and possibly more). These tables capture the cryptographic behavior of the SBox under various cryptanalytic methods (differential, linear, algebraic and boomerang analyses, respectively).

Here, every entry of the DDT, LAT, DPT or BCT is classified into one of two categories, namely positive (i.e., non-zero transition) class and negative (i.e., zero transition) class. Traditionally, the feasible transitions (positive samples) solely are used in the primary structure considers to construct the corresponding MILP model. By treating the modeling task as a binary classification problem, constraints are generated to separately capture the conditions for positive transitions and negative transitions. The primary structure, therefore, enriches the MILP framework with discriminative power, as it systematically distinguishes between valid and impossible propagations (i.e., negative samples).

Secondary Structure Moving to the secondary structure, the objective shifts from the binary classification viewpoint to integration for an SBox. Auxiliary variables are introduced to represent the state of each SBox, indicating whether it is active (participating in propagation) or inactive (not participating). Thus, the secondary structure encapsulates the primary structure’s classification into a slightly more generalized framework that tracks the activity status of each SBox.

3 Related Works

In this section, we assume that the cryptographic table from the SBox is provided (the table elucidates a binary classification problem). The objective is to formulate that table into a collection of linear constraints so that it can be used in a MILP formulation.

3.1 Representation of an SBox using Convex Hull (CH)

The *Convex Hull* (CH) of a set P of points is the smallest convex set that contains all the points in P . The process involves calculating the H-representation, which is achieved by determining the cutting planes

that collectively form the convex hull. Specifically, we apply the implementation from Sage⁵ to extract the H-representation. Consequently, the H-representation yields a system of w linear constraints, taking the form, $A \cdot [x_0, \dots, x_{n-1}, y_0, \dots, y_{n-1}]^\top \leq b^\top$. Here, A is the $w \times 2n$ matrix and $b = (b_0, b_1, b_2, \dots, b_{2n-1})$. Due to the innate implementation in Sage, both matrix A and vector b consist solely of integers. Each linear constraint serves to invalidate certain points, associated with unattainable differential propagation.

3.2 Modeling using Conditional Differential Behavior

The logical relationship $(x_0, \dots, x_{m-1}) = (\delta_0, \dots, \delta_{m-1}) \in \{0, 1\}^m \subseteq \mathbb{Z}^m$, which implies $y = \delta \in \{0, 1\} \subseteq \mathbb{Z}$, can be effectively captured through the following linear constraint: $\sum_{i=0}^{m-1} (-1)^{\delta_i} x_i + (-1)^{\delta+1} y - \delta + \sum_{i=0}^{m-1} \delta_i \geq 0$

Here, we consider (x_0, x_1, x_2, x_3) and (y_0, y_1, y_2, y_3) as the variables for input and output differences of a 4-bit SBox. Suppose there is an impossible propagation in the DDT of the SBox: $(1010) \rightarrow (0111)$, indicating that the input difference (1010) does not propagate to (0111) . Utilizing relationship noted earlier, the linear constraint that asserts the impossible point (i.e., negative sample) $(1001, 0111)$ is expressed as: $-x_0 + x_1 - x_2 + x_3 + y_0 - y_1 - y_2 - y_3 + 4 \geq 0$.

For an SBox, if there are n impossible paths evident in a DDT, at most n linear constraints are necessary to accurately model the DDT. However, it is possible to reduce the value of n by merging existing constraints, thereby generating new ones.

As an illustration, consider two impossible propagations (i.e., negative samples): $(1010) \rightarrow (0111)$ and $(1010) \rightarrow (0110)$. A combined linear constraint: $-x_0 + x_1 - x_2 + x_3 + y_0 - y_1 - y_2 \geq -3$.

effectively eliminates both of these impossible points simultaneously. This consolidation of constraints helps streamline the representation of impossible propagations and enhances the efficiency of the system.

3.3 Selecting Best Constraints from Convex Hull

When applying an MILP model, obtaining a feasible solution does not guarantee a valid differential path. The objective is to reduce the number of active SBoxes over a broader range, with the optimal value being less than or equal to the actual minimum count of active SBoxes. To achieve this goal, linear constraints are sought that can effectively prune parts of the MILP model while preserving the valid differential characteristics within the region. Various algorithms are proposed for reducing the number of constraints in an SBox representation.

Algorithm 1: Greedy Approach [34]

Input:

I_{Hull} : Constraints from convex hull of an SBox.

P_{IT} : Impossible transitions (i.e., negative samples) from an evaluation table of an SBox

Output:

I_R : Reduced set of constraints

```

1:  $I_R \leftarrow \emptyset$ 
2: while  $P_{IT} \neq \emptyset$  do
3:    $I_Q \leftarrow$  Possible constraints in  $I_{Hull}$  which maximizes number of removed impossible transitions from  $P_{IT}$ .
4:    $P_{IT} \leftarrow P_{IT} - \{\text{Removed impossible transitions using } I_Q\}$ 
5:    $I_{Hull} \leftarrow I_{Hull} - \{I_Q\}$ 
6:    $I_R \leftarrow I_R \cup \{I_Q\}$ 
7: return  $I_R$ 

```

Greedy algorithm based approach. The discrete points within the H-Representation (convex hull) often generate a substantial number of constraints. An effective strategy involves identifying the best valid constraints that maximize the elimination of impossible differential patterns within the feasible region of the convex hull. Algorithm 1, as proposed by Sun et al., presents this greedy approach.

⁵<https://www.sagemath.org/>

Algorithm 2: Modeling by selecting random set of constraints [18]

Inputs:

 P_{VT} : Input set corresponding to possible transitions in an SBox. k : Number of constraints to be counted together.

Output:

 I_R : Reduced set of constraints

```

1:  $I_{Hull} \leftarrow ConvHull(P_{VT})$ 
2:  $I_R \leftarrow H$ 
3: for all  $p \in P_{IT}$  do
4:   Find all the  $k$  subsets of constraints from  $I_{Hull}$ 
5:   Choose a subset such that  $p$  belongs to the hyperplanes of  $I_{Q_1}, I_{Q_2}, \dots, I_{Q_k}$ 
6:    $I_{Q_{new}} = I_{Q_1} + \dots + I_{Q_k}$ 
7:   if  $I_{Q_{new}}$  invalidates new impossible points other than that removes by the original constraints
8:      $I_R \leftarrow I_R \cup \{I_{Q_{new}}\}$ 
9:   end if
10: return  $I_R$ 

```

Modeling via selection of random subset of constraints. The generation of an extensive set of new constraints by randomly adding k constraints often results in redundancy, as they are likely to cover the entire space $\{0, 1\}^m$. However, when k hyperplanes from the H-representation share a vertex within the $\{0, 1\}^m$ cube, combining the corresponding constraints can potentially yield a new constraint $I_{Q_{new}}$. Importantly, this new hyperplane represented by $I_{Q_{new}}$ should intersect with the cube at least once. In such cases, $I_{Q_{new}}$ has the potential to invalidate a distinct set of impossible points within the H-representation compared to the older constraints. The complete procedure is summarized in Algorithm 2.

Algorithm 3: MILP based reduction [30]

Inputs:

 P_{IP} : Impossible patterns (i.e., negative samples) from evaluation table of an SBox. P_{VT} : Possible transitions (i.e., positive samples) from evaluation table of an SBox.

Output:

 I_R : Reduced set of constraints

```

1:  $I_{Hull} \leftarrow ConvHull(P_{VT})$ 
2:  $I_R \leftarrow I_{Hull}$ 
3: Create a table  $T$  of size  $|I_R| \times |P_{IP}|$  where  $T_{i,j} = 1$  if constraint  $I_{R_i}$  removes pattern  $P_{IP_j}$ , else set  $T_{i,j} = 0$ 
4: Set  $m = |I_R|$ 
5: Generate  $m$  binary variables  $b_1, b_2, \dots, b_m$  such that  $b_i = 1$  means constraint  $i$  is included in the solution space else  $b_i = 0$ .
6:  $constraintSet \leftarrow \emptyset$ 
7: for point  $\in P_{IP}$  do
8:    $constraintSet \leftarrow constraintSet \cup \{Construct\ a\ constraint\ b_k, \dots, b_l \geq 1\ such\ that\ p\ is\ removed\ by\ at\ least\ one\ constraint\ applying\ table\ T\}$ 
8:   ▷ Generate constraints
9: Create an objective function  $\sum_{i=1}^m b_i$  with constraints  $constraintSet$  and solve to get best constraints ( $I_R$ ) that generate a stricter feasible region after maximizing the removed impossible differential patterns.
10: return  $I_R$ 

```

MILP-based reduction algorithm. Sasaki and Todo [30] introduce a method that employs a standard MILP solver to derive a minimized set of constraints. Initially, they identify all the impossible differential points within the DDT of an SBox. These points serve as the foundation for generating impossible patterns. For each impossible pattern generated, the algorithm assesses which subset of constraints can effectively render the pattern invalid. This step pinpoints the crucial constraints required for pattern invalidation. Subsequently, they formulate a constraint specifying that each impossible pattern must be made infeasible within the valid region by at least one constraint. An MILP problem is then constructed to minimize the

overall set of constraints while adhering to the constraints established in the prior steps. The formulated MILP problem is solved, yielding a minimized set of constraints that effectively reduces the number of impossible patterns. The method is summarized in Algorithm 3.

4 Our First Idea: Switching Positive and Negative Samples

In this section, the first proposed approach is described, wherein the roles of the positive and negative samples from the given SBox evaluation table are swapped. The fundamental concept involves treating the positive samples as negative and vice versa. Thus, the complementary problem is modeled, followed by the construction of the overarching MILP problem, and from its solution, the approach reverts to the original problem (by taking the complement again). It is demonstrated that, for certain SBox evaluation tables, this method is beneficial in reducing the number of linear constraints.

To demonstrate the basic concept of this, we present a suitable example. Let $f_1(X_3, X_2, X_1, X_0)$ be a set of 4 elements in $\mathbb{F}_2^4$, defined as: $f_1 = \{0000, 0011, 1100, 1111\}$ that represents the non-zero elements.

The goal is to find a minimal representation of the set f_1 . Figure 1(a) shows the set f_1 represented in a Karnaugh map. The minimized Boolean expression after applying the Karnaugh map method (over 1's), $f_1^1(X_3, X_2, X_1, X_0) = X_3'X_2'X_1'X_0' + X_3'X_2'X_1X_0 + X_3X_2X_1'X_0' + X_3X_2X_1X_0$ (where, each X_i' denotes the complement of binary value X_i). Alternatively, the 0's may be applied to find the minimized expression (see Figure 1(b)). Thus, the minimized Boolean expression applying the Karnaugh map method (over 0's) is $f_1^0(X_3, X_2, X_1, X_0) = (X_3 + X_2')(X_3' + X_2)(X_1 + X_0')(X_1' + X_0)$. It can be observed that when using the 0's (i.e., negative samples) map, the resulting product of sum terms is simpler and contains fewer literals than the 1's (i.e., positive samples) map of product of sum.

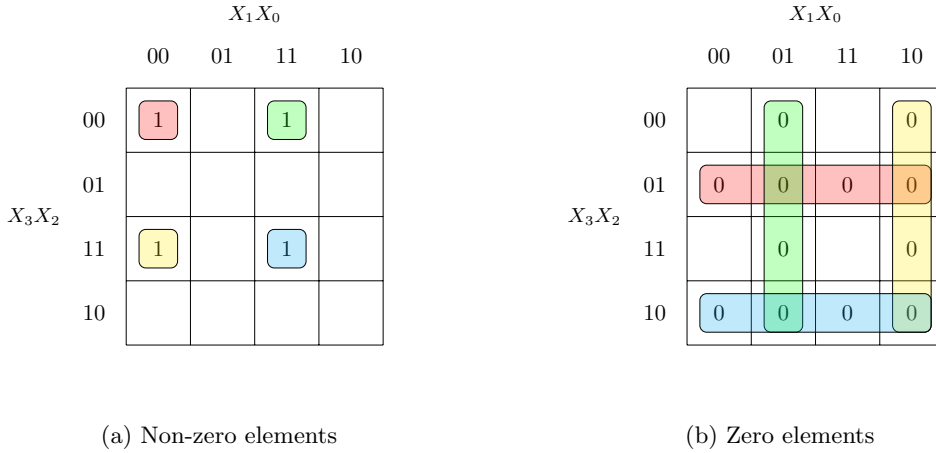
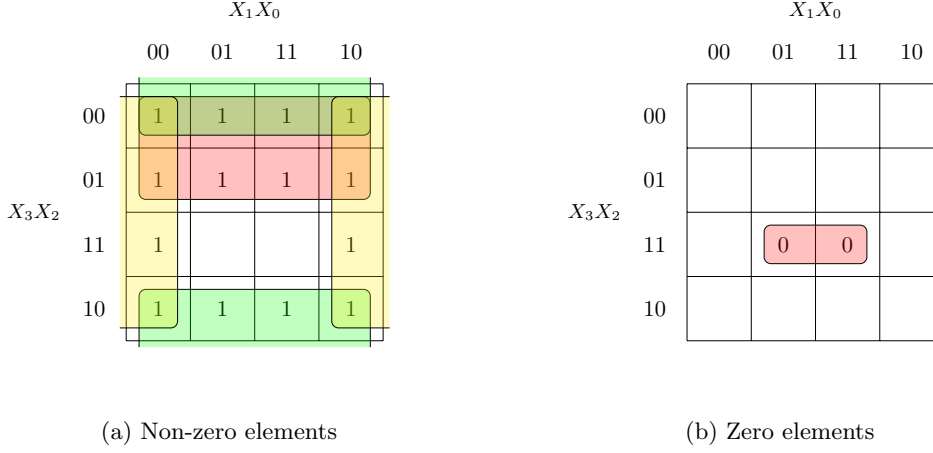


Fig. 1: K-Map for f_1

Suppose $f_2(X_3, X_2, X_1, X_0)$ be a set of 14 elements in $\mathbb{F}_2^4$ in another example, where $f_2 = \{0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1110\}$. Figure 2(a) presents the representation of the set f_2 using a Karnaugh map. By applying the Karnaugh map minimization procedure over the 1's, the simplified Boolean expression is, $f_2^1(X_3, X_2, X_1, X_0) = X_0' + X_2' + X_3'$. Alternatively, minimization can also be performed using the 0's (see Figure 2(b)), yielding, $f_2^0(X_3, X_2, X_1, X_0) = X_0' + X_2' + X_3'$. It is worth noting that minimization with respect to the 0's leads directly to a product-of-sums form, which contains the same number of literals as the product-of-sums expression derived from the 1's map. However, obtaining the minimal expression is generally more straightforward when employing the 0's map.

This approach can be extended to the modeling of SBoxes in cryptanalysis-based MILP formulations by altering the constraint-generation process derived from the LAT, DDT, DPT and BCT. Typically, in

Fig. 2: K-Map for f_2

classical SBox modeling, the binary variables representing input - output patterns are assigned a value of 1 to indicate the presence of an active bit or a valid approximation/differential, and the resulting constraints are constructed to enforce the propagation constraints. However, in certain scenarios, such as when searching for impossible differentials, zero-correlation linear approximations, or other structural properties, it is beneficial to invert this encoding, assign 0 to indicate an active or valid position, and 1 to indicate inactivity or invalid propagation. Through this approach, the constraint system is reformulated so that the constraints capture the absence of specific transitions or correlations rather than their presence. This “complementary variable encoding” can simplify the MILP model when the cryptanalytic objective is to identify patterns of non-propagation or to detect impossible transitions.

This inversion of bit semantics allows the evaluation table to derive constraints to be directly reused by simply flipping the binary mapping while preserving their structural relationship to the underlying combinatorial properties of the SBox. This approach is particularly advantageous when integrating SBox constraints into multi-round block cipher models, where representing zero-propagation conditions efficiently can reduce solver complexity and improve runtime performance.

Switching Back to Positive Samples: Modeling for Exact Problem

One may notice that the model we discuss here actually finds the formulation of the complementary problem, that is, it looks for the inactive SBoxes (as it treats the negative samples of an evaluation table as positive samples, and vice-versa). However, since our objective is to find the minimum number of active SBoxes, we need to adjust the MILP problem formulation. One idea is as follows.

After we model the negative samples as the positive samples (and vice-versa), we make the MILP problem as “maximize number of inactive SBoxes”. Then we get the maximum possible number of SBoxes that can be inactive (say, the result from the MILP solver is I). Then, we subtract that from the total number of SBoxes. Say, for a cipher with 32 SBoxes at one round and tested for 10 rounds, we have in total $32 \times 10 = 320$ SBoxes. Therefore, the minimal number of active SBoxes for our case will be $320 - I$.

Aside from that, another idea is to change the indicator variable that captures if the SBox is active or not, to reflect the complementary nature. For example, if we have an indicator variable S that is 1 when the SBox is active and 0 otherwise, we can replace it as $1 - S$. The objective function will remain the same, i.e., “minimize number of active SBoxes”. However, since we have changed the indicator variable, the MILP solver will now find the minimum number of active SBoxes directly.

Table 2: Minimum number of linear constraints for DDT
(a) Including 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	157, 99	73, 237	34, 24	31, 22	30, 21	21, 17	21, 17	21, 17
KLEIN		150, 106	158, 311	33, 22	30, 22	28, 21	23, 17	23, 17	23, 17
LILLIPUT		150, 106	183, 324	31, 27	30, 26	29, 23	26, 19	26, 20	25, 19
MIBS		150, 106	173, 378	32, 27	30, 24	29, 23	26, 20	26, 20	26, 20
MIDORI S0		159, 97	142, 239	32, 23	30, 25	27, 21	26, 16	26, 17	26, 16
MIDORI S1		150, 106	166, 367	29, 26	28, 24	28, 22	26, 20	26, 20	26, 20
MINALPHER		150, 106	140, 338	26, 24	26, 24	26, 22	21, 19	21, 19	21, 18
PICCOLO		159, 97	72, 202	27, 23	26, 24	21, 21	21, 16	21, 16	21, 16
PRESENT		159, 97	115, 237	35, 22	32, 22	30, 21	21, 17	21, 17	20, 17
PRIDE		159, 97	74, 194	30, 22	28, 22	26, 21	21, 16	21, 17	21, 17
PRINCE		150, 106	151, 300	31, 26	31, 26	29, 22	25, 19	25, 19	25, 18
RECTANGLE		159, 97	70, 267	25, 25	24, 23	24, 21	19, 17	20, 17	20, 16
SKINNY		159, 97	72, 202	28, 24	27, 24	26, 21	21, 16	21, 16	21, 16
TWINE		150, 106	183, 324	33, 23	30, 25	29, 23	26, 19	26, 20	25, 19
ASCON	5	707, 317	216, 2415	66, 51	64, 48	63, 40	48, 32	48, 31	47, 31
FIDES-5		527, 497	768, 910	101, 96	96, 90	84, 79	64, 61	64, 64	63, 62
KECCAK		707, 314	182, 316	56, 50	52, 47	52, 46	37, 34	37, 36	32, 36
ICEPOLE		687, -	347, -	80, -	77, -	72, -	57, -	59, -	57, -
SC2000-5		527, 497	800, 908	98, 99	95, 92	84, 82	66, 64	66, 65	65, 63
FIDES-6	6	-, 2017	-, 223	-, -	-, -	-, -	-, 180	-, 184	-, -
SC2000-6		-, 1954	-, 241	-, -	-, -	-, -	-, 214	-, 89	-, -

*: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

(b) Excluding 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	98, 127	119, 141	27, 26	27, 24	25, 22	18, 17	18, 17	18, 17
KLEIN		120, 105	208, 281	29, 23	27, 22	25, 20	21, 18	21, 18	21, 18
LILLIPUT		120, 105	225, 297	28, 23	27, 22	24, 21	20, 19	20, 19	20, 19
MIBS		120, 105	213, 320	28, 26	27, 25	25, 23	22, 20	22, 20	21, 20
MIDORI S0		129, 96	164, 217	33, 23	30, 22	27, 20	24, 17	24, 17	24, 17
MIDORI S1		120, 105	209, 334	26, 25	25, 25	23, 22	21, 20	21, 20	21, 20
MINALPHER		120, 105	214, 320	27, 24	26, 24	25, 22	19, 19	19, 18	19, 18
PICCOLO		129, 96	106, 186	25, 23	25, 22	24, 20	19, 15	19, 16	19, 15
PRESENT		129, 96	170, 233	32, 21	30, 20	26, 18	18, 17	18, 15	18, 15
PRIDE		129, 96	107, 170	28, 21	27, 21	25, 21	20, 17	20, 19	20, 19
PRINCE		120, 105	202, 270	27, 26	27, 25	25, 22	22, 18	22, 19	22, 18
RECTANGLE		129, 96	104, 158	24, 24	22, 24	20, 22	16, 16	16, 17	16, 17
SKINNY		129, 96	106, 186	28, 24	27, 23	24, 20	19, 15	19, 16	19, 15
TWINE		120, 105	225, 297	28, 25	27, 24	24, 21	20, 19	20, 19	20, 19
ASCON	5	645, 316	273, 1739	65, 48	63, 45	59, 40	46, -	46, 31	45, -
FIDES-5		465, 496	1034, 729	93, 95	84, 87	78, 80	61, -	61, -	59, -
KECCAK		645, 316	229, 308	52, 53	50, 49	47, 45	33, -	32, -	31, -
SC2000-5		465, 496	1091, 747	92, 95	87, 90	76, 82	61, -	61, -	59, -
FIDES-6	6	1953, 2016	8200, 7012	257, 275	248, 263	-, -	-, -	-, -	-, -
SC2000-6		2016, 1953	9816, 11238	292, 294	286, 284	-, -	-, -	-, -	-, -

*: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

Table 3: Minimum number of linear constraints for (absolute) LAT
(a) Including 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	123, 133	50, 211	19, 18	19, 18	19, 16	15, 13	15, 14	15, 13
KLEIN		105, 151	285, 133	27, 31	27, 29	24, 27	18, 23	18, 23	18, 23
LILLIPUT		105, 151	239, 126	26, 32	26, 32	24, 32	19, 22	19, 22	18, 22
MIBS		105, 151	347, 124	28, 36	26, 35	23, 35	19, 26	19, 26	19, 26
MIDORI S0		123, 133	195, 92	31, 28	29, 28	25, 25	19, 16	19, 18	19, 17
MIDORI S1		105, 151	338, 104	25, 32	25, 31	23, 29	18, 22	18, 22	18, 22
MINALPHER		105, 151	304, 143	25, 32	25, 31	21, 28	18, 23	18, 23	18, 23
PICCOLO		123, 133	64, 76	22, 17	21, 16	20, 16	14, 11	14, 11	14, 11
PRESENT		123, 133	169, 137	27, 29	25, 26	23, 24	17, 13	17, 17	17, 17
PRIDE		123, 133	67, 72	21, 16	21, 16	21, 15	15, 11	15, 12	15, 11
PRINCE		105, 151	298, 114	25, 32	25, 31	23, 29	18, 20	18, 21	18, 20
RECTANGLE		123, 133	55, 186	22, 17	22, 16	21, 15	18, 12	19, 12	18, 12
SKINNY		123, 133	64, 76	21, 16	21, 16	20, 16	14, 11	14, 11	14, 11
TWINE		105, 151	239, 126	27, 32	26, 32	24, 32	19, 22	19, 22	18, 22
ASCON	5	647, 377	304, 1613	83, 70	79, 69	75, 58	-, 44	52, 44	50, -
FIDES-5		527, 497	765, 889	94, 96	94, 90	83, 77	-, 62	66, 63	64, 62
ICEPOLE		497, -	857, -	102, -	102, -	90, -	80, 64	84, -	81, -
KECCAK		647, 377	137, 236	69, 67	67, 66	67, 65	37, 38	42, 40	37, 39
SC2000-5	6	527, 497	801, 917	100, 93	99, 88	84, 78	68, 63	68, 64	67, 63
FIDES-6		-, 3025	-, 2699	-, 185	-, 177	-, 159	-, -	-, 119	-, -
SC2000-6		-, 3214	-, 1053	-, 320	-, 315	-, 307	-, -	-, 274	-, -

★: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

(b) Excluding 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	93, 132	74, 116	18, 19	18, 18	17, 15	13, 12	13, 13	13, 12
KLEIN		75, 150	413, 92	23, 28	22, 27	19, 27	15, 23	-, 23	-, 23
LILLIPUT		75, 150	376, 87	19, 31	19, 31	18, 31	15, 22	15, 21	15, 21
MIBS		75, 150	497, 86	23, 35	22, 35	19, 35	16, 26	16, 25	16, 25
MIDORI S0		93, 132	248, 82	25, 25	23, 25	20, 25	16, 16	17, 16	16, 17
MIDORI S1		75, 150	449, 76	24, 30	22, 30	19, 29	15, 22	15, 22	15, 22
MINALPHER		75, 150	438, 100	23, 29	21, 28	19, 27	16, 23	16, 22	16, 22
PICCOLO		93, 132	103, 62	19, 15	18, 15	17, 15	12, 11	12, 11	11, 10
PRESENT		93, 132	276, 93	20, 28	19, 24	18, 16	15, 16	15, 17	15, 17
PRIDE		93, 132	110, 59	19, 13	19, 13	18, 13	12, 10	12, 9	11, 9
PRINCE		75, 150	379, 84	22, 30	21, 30	18, 29	15, 19	15, 19	15, 10
RECTANGLE		93, 132	84, 90	21, 19	21, 18	20, 16	13, 12	14, 12	13, 11
SKINNY		93, 132	103, 62	19, 15	18, 15	17, 15	12, 11	12, 11	11, 10
TWINE		75, 150	376, 86	20, 32	19, 32	18, 31	15, 22	15, 21	15, 21
ASCON	5	585, 376	390, 1107	76, 68	73, 65	69, 60	-, -	50, 45	47, -
FIDES-5		465, 495	1034, 716	90, 91	87, 84	76, 77	-, -	62, -	61, -
KECCAK		585, 376	164, 228	66, 66	64, 65	62, 64	-, -	41, -	36, -
SC2000-5		465, 496	1085, 760	92, 89	87, 87	77, 79	-, -	63, -	62, -
FIDES-6	6	945, 3024	17147, 2392	115, 182	110, 172	-, 159	-, -	-, -	-, -
SC2000-6		-, -	-, 942	-, 307	-, -	-, -	-, -	-, -	-, -

★: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

5 Our Second Idea: Exclusion of Trivial Transitions in SBox Modeling

In the construction of DDT, LAT, DPT and BCT; the trivial zero transitions are included. These entries correspond to cases where at least one of the input and output characteristics is zero. Since the goal of SBox modeling is to capture meaningful propagation patterns that reflect the cryptographic strength or weakness of the design, such trivial transitions can be considered redundant.

By deliberately excluding these trivial transitions, the modeling process becomes more efficient. Specifically, the MILP representation, which relies on generating sets of constraints for feasible and infeasible transitions, can be significantly simplified. The exclusion of trivial cases results in a direct reduction of the number of constraints required, since no modeling effort is spent on transitions that do not affect the differential, linear, division, or boomerang propagation properties of the cipher. This reduction not only decreases computational overhead but also improves solver performance when searching for optimal or impossible trails.

Moreover, this selective elimination aligns with the broader principle of focusing on cryptographically significant events in the analysis. By ignoring non-contributing transitions, the SBox model preserves all essential information needed for security evaluation while discarding unnecessary complexity. The gain here is twofold: a reduction in the mathematical burden of constraint generation and an improvement in the clarity of the cryptographic structure being analyzed. Thus, ignoring trivial transitions is not merely a simplification, but a targeted optimization that enhances both the efficiency and precision of SBox modeling.

6 Results

We have the approach to all the earlier approaches for SBox modeling. To make the explanation clear, we consider both types of constraints generated from zero and non-zero transitions in LAT/DDT/DPT/BCT. Our results section has mainly two parts, each having four tables. The first part (Table 2(a), 3(a), 4(a), 5(a)) provides the number of constraints for different SBox evaluation tables. The second part (Table 2(b), 3(b), 4(b), 5(b)) explains the number of constraints for different SBox evaluation tables excluding the first row and first column, that is, ignoring the trivial transitions. Each resultant table describes the number of constraints required for modeling different 4, 5 and 6-bit SBoxes applying different SBox modeling approaches. Each cell of any table has one pair of values, namely the number of constraints required for applying negative sampling and positive sampling (in this order). For those we do not get a result, we mark by the ‘-’ symbol.

Our analysis yields superior results for a wide range of ciphers by utilizing the zero transition property detailed in the DDT Table 2(a). Specifically, applying this to the Sage approach led to improvements for almost all SBoxes tested. Additionally, we recorded advancements for KECCAK using the subset addition method [29] and for SC2000-5 in relation to the Sun et al.’s approach [34].

DDT excluding trivial transitions. Table 2(b) compares the constraints generated from zero and non-zero transitions (i.e., positive and negative samples) in DDT for different modeling, ignoring the trivial transitions. This approach leads to significant improvements for almost every existing approach for a wide range of ciphers, including RECTANGLE, FIDES-5, KECCAK, SC2000-5, FIDES-6 and SC2000-6 when applied to the DDT as given in Table 2(b).

LAT. Table 3(a) presents a comparison of different SBox modeling algorithms for LAT under both positive and negative sampling. In this case, we observe improved results for SBoxes used in KLEIN, LILLIPUT, MIBS, MIDORI S1, MINALPHER, PRINCE and TWINE across most of the modeling approaches.

LAT excluding trivial transitions. Table 3(b) compares the constraints generated from zero and non-zero transitions in LAT for different modeling, ignoring the trivial transitions. Here we obtain improved results for SBoxes like KLEIN, LILLIPUT, MIBS, MIDORI S1, MINALPHER, PRESENT, PRINCE, TWINE and FIDES-6 for most of the modeling algorithms.

Table 4: Minimum number of linear constraints for DPT
(a) Including 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	208, 48	50, 213	31, 14	31, 14	31, 13	23, 9	23, 9	23, 9
KLEIN		206, 50	50, 117	21, 11	20, 10	20, 9	15, 7	15, 7	13, 6
LILLIPUT		210, 46	49, 122	20, 11	20, 11	20, 9	14, 7	14, 7	12, 7
MIBS		204, 52	51, 84	20, 9	19, 9	19, 8	14, 7	14, 7	12, 7
MIDORI S0		209, 47	50, 45	19, 6	19, 6	19, 6	14, 5	14, 5	12, 5
MIDORI S1		208, 48	47, 75	20, 10	19, 9	19, 9	13, 7	13, 7	12, 7
MINALPHER		206, 50	50, 134	21, 11	20, 10	20, 9	15, 7	15, 7	14, 7
PICCOLO		211, 45	47, 98	27, 13	26, 13	26, 12	20, 8	20, 8	20, 7
PRESENT		210, 46	48, 103	22, 10	20, 10	20, 9	15, 7	15, 7	13, 7
PRIDE		213, 43	47, 108	25, 11	24, 11	24, 10	18, 7	18, 7	17, 6
PRINCE		207, 49	48, 86	20, 12	20, 11	20, 10	14, 8	14, 9	12, 8
RECTANGLE		208, 48	52, 197	31, 15	31, 14	31, 13	24, 9	24, 9	23, 9
SKINNY		211, 45	47, 98	27, 13	26, 13	26, 12	20, 8	20, 8	20, 7
TWINE		210, 46	49, 122	21, 11	20, 10	20, 9	14, 7	14, 7	12, 7
ASCON	5	835, 189	158, 1393	99, 22	99, 22	99, 20	72, 13	72, 15	67, 15
FIDES-5		825, 199	173, 2012	87, 23	83, 23	81, 20	56, -	56, -	54, -
KECCAK		888, 136	111, 412	91, 21	91, 19	91, 17	61, -	61, 15	61, 15
ICEPOLE		873, 151	121, 18	31, 8	31, 8	26, 8	16, -	16, 7	16, 6
SC2000-5	6	883, 141	174, 1719	68, 24	67, 24	65, 19	60, -	60, -	55, -
FIDES-6		3678, -	403, -	178, -	176, -	173, -	-, -	152, -	149, -
SC2000-6		3734, -	204, -	51, -	49, -	45, -	-, -	33, -	28, -

★: Our implementation of Boura and Coggia's method [18]

Each cell indicates (negative samples, positive samples)

(b) Excluding 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	177, 48	49, 213	21, 14	21, 14	21, 12	16, 8	16, 8	17, 8
KLEIN		175, 50	68, 117	8, 9	8, 8	7, 7	6, 6	6, 6	6, 5
LILLIPUT		179, 46	54, 122	11, 10	10, 9	9, 8	7, 6	7, 6	7, 6
MIBS		173, 52	62, 84	8, 7	8, 7	7, 7	6, 6	6, 6	5, 6
MIDORI S0		178, 47	49, 45	9, 5	9, 5	8, 5	7, 4	7, 4	7, 4
MIDORI S1		177, 48	51, 75	8, 9	8, -	7, 8	6, 6	6, 6	6, 6
MINALPHER		175, 50	56, 134	10, 9	10, 9	9, 8	7, 6	7, 6	6, 6
PICCOLO		180, 45	69, 98	17, 12	17, 12	16, 11	14, 7	14, 7	13, 7
PRESENT		179, 46	48, 103	9, 9	9, 9	9, 8	7, 6	7, 6	6, 6
PRIDE		182, 43	77, 108	13, 10	13, 10	13, 9	10, 7	10, 7	10, 6
PRINCE		176, 49	50, 86	9, 10	9, 10	8, 9	7, 7	7, 7	7, 7
RECTANGLE		177, 48	52, 197	21, 14	21, 13	21, 11	16, 8	16, 9	16, 8
SKINNY		180, 45	69, 98	17, 13	17, 12	16, 11	14, 7	14, 7	13, 7
TWINE		179, 46	54, 122	11, 10	11, 9	9, 8	7, 6	7, 6	7, 6
ASCON	5	772, 189	178, 1393	86, 22	82, 20	81, 18	-, -	58, 15	53, -
FIDES-5		762, 199	171, 2012	73, 21	69, 22	65, 18	-, -	47, -	45, -
KECCAK		825, 136	128, 412	81, 19	81, 18	81, 15	-, -	56, -	56, -
SC2000-5		820, 141	217, 1719	47, 23	47, 23	46, 18	-, -	45, -	45, -
FIDES-6	6	3551, 418	763, -	148, -	144, -	144, -	-, -	132, -	132, -
SC2000-6		3607, 362	174, -	17, -	17, -	16, -	-, -	-, -	-, -

★: Our implementation of Boura and Coggia's method [18]

Each cell indicates (negative samples, positive samples)

Table 5: Minimum number of linear constraints for BCT
(a) Including 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	102, 154	173, 104	25, 26	24, 26	23, 25	17, 19	17, 17	16, 17
KLEIN		122, 144	246, 164	30, 31	28, 27	24, 25	20, 23	20, 19	20, 19
LILLIPUT		120, 136	225, 210	29, 29	28, 28	26, 26	21, 22	21, 19	21, 19
MIBS		120, 136	213, 220	29, 30	29, 28	26, 26	22, 23	22, 20	21, 20
MIDORI SO		107, 149	127, 146	26, 25	24, 24	23, 22	18, 18	18, 16	17, 15
MIDORI S1		105, 151	242, 159	25, 33	24, 32	22, 20	18, 28	20, 24	18, 24
MINALPHER		122, 144	215, 182	30, 30	29, 30	27, 20	21, 26	21, 22	21, 22
PICCOLO		103, 153	139, 83	24, 23	23, 22	21, 22	16, 16	16, 14	15, 14
PRESENT		107, 149	239, 147	26, 30	26, 28	23, 28	18, 24	18, 21	17, 20
PRIDE		103, 153	115, 70	25, 26	23, 25	22, 24	16, 16	16, 16	16, 16
PRINCE		105, 151	269, 159	27, 31	25, 31	23, 29	19, 23	19, 21	19, 20
RECTANGLE		107, 149	162, 103	26, 25	25, 23	21, 22	15, 18	15, 16	15, 16
SKINNY		103, 153	139, 83	23, 22	23, 22	21, 22	16, 16	16, 14	-, 14
TWINE		120, 136	225, 210	30, 29	28, 27	26, 26	21, 23	21, 19	21, 19
ASCON	5	445, 579	1154, 666	57, 64	54, 60	47, 54	-, 42	38, 40	-, 38
FIDES-5		465, 559	1034, 815	94, 97	93, 93	78, 83	61, 66	-, 63	-, 62
KECCAK		445, 579	1230, 435	40, 58	41, 56	36, 55	-, 41	-, 45	-, 43
SC2000-5		465, 559	1091, 831	94, 100	92, 99	77, 84	-, 70	-, 65	-, 64
FIDES-6	6	1953, 2143	8200, 6290	260, 286	-, 278	-, -	-, -	-, -	-, -
SC2000-6		1980, 2206	12384, 7158	278, 314	-, 305	-, -	-, -	-, -	-, -

★: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

(b) Excluding 0th row and 0th column

SBox	Size	Naïve [4]	Full Convex Hull [34]	Chosen Convex Hull (Deterministic) [34]	Chosen Convex Hull (Randomized) [4, 28]	Sasaki, Todo [30]	Boura, Coggia* [18]	Subset Addition [29] $k = 2$	Subset Addition [29] $k = 3$
GIFT	4	102, 123	173, 143	24, 21	21, 21	21, 21	17, 17	17, 17	16, 17
KLEIN		112, 113	246, 226	28, 25	25, 24	23, 23	20, 19	20, 19	20, 19
LILLIPUT		120, 105	225, 297	26, 25	23, 23	21, 21	20, 19	20, 19	20, 19
MIBS		120, 105	213, 320	29, 26	26, 25	23, 23	22, 20	22, 20	21, 20
MIDORI SO		107, 118	127, 162	25, 23	20, 20	20, 20	17, 16	17, 16	17, 15
MIDORI S1		105, 120	242, 220	22, 29	29, 28	27, 27	17, 24	18, 24	17, 24
MINALPHER		112, 113	215, 249	29, 26	27, 26	25, 25	20, 22	20, 22	20, 22
PICCOLO		103, 122	139, 86	22, 24	23, 24	21, 21	15, 14	15, 14	15, 14
PRESENT		107, 118	239, 180	25, 27	28, 27	24, 24	18, 21	18, 21	17, 20
PRIDE		103, 122	115, 88	22, 25	22, 22	22, 22	16, 14	16, 16	15, 16
PRINCE		105, 120	269, 221	25, 29	28, 27	25, 25	19, 21	19, 21	19, 20
RECTANGLE		107, 118	162, 126	24, 22	21, 21	19, 19	15, 16	15, 16	15, 16
SKINNY		103, 122	139, 86	22, 24	24, 24	21, 21	15, 14	15, 14	15, 14
TWINE		120, 105	225, 297	28, 25	24, 23	21, 21	20, 19	20, 19	20, 19
ASCON	5	445, 516	1154, 712	54, 60	57, 56	50, 50	-, -	37, 40	35, 38
FIDES-5		465, 496	1034, 729	94, 94	87, 89	80, 80	-, -	61, 63	59, 62
KECCAK		445, 516	1230, 552	40, 64	65, 64	55, 55	-, -	32, 45	-, 43
SC2000-5		465, 496	1091, 747	93, 96	89, 91	82, 82	-, -	-, 65	-, 64
FIDES-6	6	1953, 2016	8200, 7012	256, 269	269, -	-, -	-, -	-, -	-, -
SC2000-6		1890, 2079	12384, 8568	275, -	296, -	-, -	-, -	-, -	-, -

★: Our implementation of Boura and Coggia’s method [18]

Each cell indicates (negative samples, positive samples)

DPT. Table 4(a) presents a comparative analysis of different S-box modeling techniques for DPT under both positive and negative sampling strategies. The results indicate that improvements are observed primarily in the Sage generated outcomes, while across all evaluated ciphers, the negative sampling approach consistently demonstrates superior performance.

DPT excluding trivial transitions. Table 4(b) compares the constraints derived from zero and non-zero transitions in the DPT across different modeling approaches, excluding the trivial transitions. In this setting, improvements occur only in the Sage-generated results, while for most ciphers the negative sampling strategy yields better performance. In particular, PRINCE shows strong results when evaluated with the greedy [34], randomized greedy [29] and MILP-aided [30] approaches.

BCT. Table 5(a) describes the comparisons of different SBox modeling algorithms for BCT against positive and negative sampling. Here we obtain improved results for SBoxes like GIFT, KLEIN, MIBS, MIDORI S0, MIDORI S1, MINALPHER, PRESENT, PRINCE, PRIDE, ASCON, FIDES-5 and KECCAK for some of the modeling algorithms. Here, all the algorithms provide better results with respect to negative sampling.

BCT excluding trivial transitions. Table 5(b) compares the constraints generated from zero and non-zero transitions in BCT for different modeling, ignoring the trivial transitions. Here we obtain improved results for SBoxes like GIFT, KLEIN, MIBS, MIDORI S0, MIDORI S1, MINALPHER, PRESENT, PRINCE, PRIDE, ASCON, FIDES-5 and KECCAK for some of the modeling algorithms. Here, greedy [34], Boura and Coggia’s method [18], subset addition [29] algorithms have better performance with respect to negative sampling.

Overall Assessment. In order to see the impact of our work, one may compare, for instance, the results on the GIFT (absolute) LAT (from Table 3(a)) indicate that the best-known result prior to our work was 13 constraints (using [18] and [29, $k = 3$]). In contrast, our approach reduces this to 12 constraints, as can be seen from Table 3(b). Basically, we augment with 3 more options, so that the user can choose the best one out of 4.

7 Conclusion

In this paper, two efficient strategies for modeling SBoxes in MILP-based cryptanalysis are introduced. Instead of modeling possible transitions, a model based on the impossible transitions (i.e., negative samples) is created. The approach is further streamlined by excluding the redundant top row and left column (that correspond to trivial zero-to-zero transition).

The experimental results, presented across a wide range of 4, 5, and 6-bit SBoxes and various property tables including the DDT, LAT, DPT and BCT, confirm that these methods successfully produce a smaller set of linear constraints than traditional modeling approaches. By reducing the number of constraints, these models are expected to improve the efficiency of automated cryptanalysis, ultimately providing a more powerful tool for the security evaluation of modern ciphers.

A Exemplary Constraints for Evaluation Tables

For better clarity, some examples of the generated linear constraints are presented here. All variables are binary, and x ’s (respectively, y ’s) denote the input (respectively, output) characteristics.

Notice that Examples 2, 3, 4, 5 and 6 correspond to the primary structure; and Examples 7 and 8 correspond to the secondary structure (see Section 2.7 for the relevant details). The switching of positive and negative samples is handled in the secondary structure setting (it cannot be captured in the primary structure), and similarly the exclusion of zero-to-zero transitions are automatically managed by the auxiliary constraints used in the secondary structure.

Example 2 (Sun et al.’s [34] ABC SBox (DDT) with positive samples using convex hull (Sage)).

```

1 +1x2 +0x1 -2x0 +1y2 +2y1 +1y0 >= 0
2 +1x2 +2x1 -2x0 +1y2 +0y1 +1y0 >= 0
3 +1x2 +0x1 +2x0 -1y2 -2y1 -1y0 >= 2
4 +3x2 +2x1 +4x0 -1y2 -2y1 -1y0 >= 0
5 +0x2 +0x1 +0x0 +0y2 +1y1 +0y0 >= 0
6 +0x2 -1x1 +0x0 +0y2 +0y1 +0y0 >= 1
7 +1x2 +0x1 +0x0 -1y2 +0y1 +1y0 >= 0
8 -1x2 +0x1 +0x0 -1y2 +0y1 -1y0 >= 2
9 +0x2 +0x1 +1x0 +0y2 +0y1 +0y0 >= 0
10 +1x2 -2x1 +0x0 +1y2 +2y1 +1y0 >= 0

```

```

11 +1x2 +2x1 +2x0 -1y2 +0y1 -1y0 >= 0
12 +0x2 +0x1 -1x0 +0y2 +0y1 +0y0 >= 1
13 +3x2 -2x1 -2x0 +1y2 +4y1 +1y0 >= 0
14 -1x2 +0x1 +0x0 +1y2 +0y1 +1y0 >= 0
15 +0x2 +1x1 +0x0 +0y2 +0y1 +0y0 >= 0
16 +1x2 +2x1 +0x0 +1y2 -2y1 +1y0 >= 0
17 +1x2 +0x1 +0x0 +1y2 +0y1 -1y0 >= 0
18 +0x2 +0x1 +0x0 +0y2 -1y1 +0y0 >= 1
19 +1x2 -1x1 -1x0 +0y2 +1y1 +0y0 >= 1

```

□

Example 3 (Greedy Approach [34] on LILLIPUT (LAT without 0^{th} row and 0^{th} column): Negative samples).

```

1 +2x3 +1x2 +7x1 -5x0 +3y3 +4y2 -1y1 -7y0 >= 6
2 +2x3 +4x2 +1x1 +4x0 -2y3 -1y2 -3y1 +1y0 >= 1
3 -3x3 -1x2 -1x1 -1x0 +3y3 +1y2 +2y1 +2y0 >= 2
4 +1x3 -2x2 +1x1 +2x0 -3y3 -1y2 +1y1 -1y0 >= 4
5 -1x3 +2x2 -5x1 +1x0 +2y3 +5y2 -2y1 +3y0 >= 3
6 +4x3 -1x2 +1x1 -3x0 +1y3 +5y2 +6y1 +3y0 >= -2
7 -2x3 +1x2 +4x1 +2x0 +3y3 -4y2 -1y1 +1y0 >= 3
8 -4x3 -1x2 -5x1 -2x0 +4y3 -2y2 +3y1 -1y0 >= 10
9 +4x3 +1x2 -2x1 +2x0 +3y3 -1y2 -3y1 +1y0 >= 2
10 +2x3 -1x2 -2x1 -1x0 -2y3 +0y2 +1y1 +1y0 >= 4

```

```

11 -1x3 +2x2 +1x1 -2x0 -1y3 -2y2 +1y1 +1y0 >= 4
12 +2x3 +1x2 +4x1 +1x0 -3y3 +3y2 +1y1 -1y0 >= 0
13 -2x3 -1x2 -1x1 +1x0 -2y3 -1y2 -1y1 +2y0 >= 6
14 -2x3 -1x2 -2x1 +2x0 +1y3 +3y2 -2y1 +1y0 >= 4
15 +0x3 -1x2 +1x1 +0x0 -1y3 -1y2 +0y1 -1y0 >= 3
16 -2x3 +2x2 -1x1 +1x0 -2y3 +1y2 +2y1 -1y0 >= 4
17 +1x3 +2x2 -2x1 +0x0 -1y3 -1y2 -2y1 -2y0 >= 6
18 +1x3 -1x2 -1x1 -2x0 +2y3 +1y2 +3y1 -2y0 >= 3
19 -1x3 -1x2 +1x1 -2x0 -1y3 +1y2 -2y1 -1y0 >= 6

```

□

Example 4 (Randomized greedy approach [4, 29] on PRINCE (DPT without 0^{th} row and 0^{th} column): Negative samples).

```

1 +1x3 +1x2 +1x1 +2x0 +2y3 +2y2 +2y1 +3y0 >= -5
2 +1x3 +0x2 +0x1 +0x0 +1y3 +2y2 +1y1 +1y0 >= -2
3 +0x3 +1x2 +0x1 +0x0 +1y3 +1y2 +2y1 +2y0 >= -2
4 +1x3 +1x2 +2x1 +1x0 +2y3 +4y2 +4y1 +3y0 >= -6
5 +0x3 +0x2 +0x1 +1x0 +2y3 +1y2 +1y1 +2y0 >= -2

```

```

6 -1x3 -1x2 -1x1 +1x0 -1y3 +1y2 +1y1 -1y0 >= 4
7 -1x3 -1x2 +1x1 -1x0 +1y3 +1y2 -1y1 -1y0 >= 4
8 -1x3 -1x2 +1x1 -1x0 +1y3 -1y2 +1y1 -1y0 >= 4
9 -1x3 -1x2 -1x1 -1x0 -1y3 -1y2 -1y1 -1y0 >= 7

```

□

Example 5 (MILP [30] on RECTANGLE (DDT without 0^{th} row and 0^{th} column): Negative samples).

```

1 +1x3 +1x2 +0x1 +2x0 -1y3 -2y2 -1y1 -2y0 >= 4
2 +0x3 +1x2 +1x1 -1x0 +0y3 -1y2 -1y1 +1y0 >= 2
3 +1x3 -1x2 +0x1 -1x0 +1y3 -1y2 +1y1 +0y0 >= 2
4 +1x3 +1x2 +2x1 +1x0 +0y3 +0y2 -1y1 -1y0 >= 0
5 -2x3 -1x2 -2x1 -1x0 +2y3 +1y2 -1y1 -1y0 >= 6
6 +1x3 -1x2 -1x1 +1x0 +2y3 +1y2 +1y1 +1y0 >= 0
7 -1x3 +0x2 +0x1 -1x0 +1y3 +1y2 +1y1 +0y0 >= 1
8 -1x3 -1x2 +1x1 +0x0 -1y3 +0y2 -1y1 +1y0 >= 3
9 -1x3 +1x2 +1x1 +0x0 +0y3 +0y2 +1y1 -1y0 >= 1
10 -1x3 +0x2 -1x1 +1x0 +1y3 -1y2 +0y1 -1y0 >= 3

```

```

11 +0x3 +1x2 -1x1 +0x0 +1y3 -1y2 -1y1 +1y0 >= 2
12 +2x3 -1x2 -1x1 +1x0 -2y3 -1y2 -1y1 -2y0 >= 6
13 +0x3 -1x2 -1x1 +0x0 +2y3 +1y2 +1y1 +2y0 >= 0
14 -1x3 -1x2 +1x1 +1x0 +1y3 +1y2 +2y1 +2y0 >= 0
15 -1x3 -1x2 +1x1 -2x0 -1y3 -1y2 +2y1 -2y0 >= 6
16 +0x3 +2x2 -2x1 +0x0 -1y3 +1y2 +1y1 +2y0 >= 1
17 +1x3 -2x2 +1x1 -2x0 -1y3 +1y2 +2y1 -1y0 >= 4
18 +0x3 +0x2 -1x1 +0x0 -1y3 -1y2 +1y1 +1y0 >= 2
19 -1x3 -1x2 -1x1 +1x0 -1y3 +1y2 +1y1 -1y0 >= 4
20 +1x3 +1x2 -1x1 -2x0 -1y3 +2y2 -1y1 -1y0 >= 4

```

□

Example 6 (Subset Addition ($k = 2$) [29] on MIDORI S1 (BCT without 0^{th} row and 0^{th} column): Negative samples).

```

1 -5x3 +3x2 -2x1 -1x0 -5y3 +3y2 -2y1 -1y0 >= 11
2 -2x3 -1x2 +3x1 -2x0 +3y3 +1y2 -2y1 -3y0 >= 7
3 +3x3 +3x2 -1x1 -3x0 +2y3 +1y2 -2y1 +2y0 >= 3
4 +3x3 +3x2 +3x1 +2x0 +2y3 +2y2 +4y1 +3y0 >= -7
5 -3x3 +2x2 +1x1 -3x0 -3y3 +2y2 +1y1 -3y0 >= 9
6 +3x3 +1x2 -2x1 -3x0 -2y3 -1y2 +3y1 -2y0 >= 7
7 -1x3 -2x2 +3x1 -3x0 -3y3 -3y2 -1y1 +2y0 >= 10
8 +2x3 -3x2 +1x1 -3x0 -4y3 -5y2 -2y1 -1y0 >= 14
9 +3x3 +2x2 +1x1 -4x0 +5y3 -2y2 -4y1 +4y0 >= 5

```

```

10 +1x3 +2x2 -1x1 +2x0 -1y3 -1y2 +2y1 +3y0 >= 0
11 +5x3 -2x2 -4x1 +4x0 +3y3 +2y2 +1y1 -4y0 >= 5
12 -4x3 +2x2 -3x1 +1x0 -3y3 +3y2 -1y1 -2y0 >= 9
13 -1x3 -2x2 -2x1 -1x0 +2y3 -3y2 -2y1 -3y0 >= 11
14 -1x3 -4x2 -2x1 -5x0 -2y3 -3y2 -3y1 -5y0 >= 20
15 -1x3 -1x2 +2x1 +3x0 +1y3 +2y2 -1y1 +2y0 >= 0
16 -2x3 -3x2 +1x1 +3x0 -1y3 -3y2 +3y1 -2y0 >= 8
17 -3x3 -4x2 +1x1 -2x0 +5y3 -1y2 +2y1 -2y0 >= 8
18 -1x3 -2x2 -2x1 +2x0 -1y3 -2y2 -2y1 +2y0 >= 8

```

□

Example 7 (Positive sample handling in secondary structure). In this case, suppose S is a binary variable which indicates if the SBox is active (with respect to positive samples). We present two methods to handle positive samples in secondary structure, those are based on the models from [34] and [4], respectively.

The objective function is to maximize S . For constructing the secondary structure, we start from the primary structure constraints, and then modify depending on the chosen method. We start with the primary structure constraints, then modify accordingly to convert to the secondary structure.

Primary structure:

$$\begin{array}{ll}
1 & -2x_3 + 1x_2 + 0x_1 - 2x_0 + 2y_3 + 2y_2 + 1y_1 - 1y_0 \geq 3 \\
2 & +2x_3 + 0x_2 + 1x_1 - 1x_0 + 2y_3 - 2y_2 + 2y_1 - 2y_0 \geq 3 \\
3 & +1x_3 + 3x_2 + 2x_1 - 3x_0 - 2y_3 + 2y_2 + 2y_1 + 3y_0 \geq 1 \\
4 & +1x_3 + 3x_2 + 3x_1 + 2x_0 - 1y_3 + 0y_2 - 2y_1 - 2y_0 \geq 2 \\
5 & -1x_3 + 2x_2 + 0x_1 - 2x_0 + 2y_3 - 2y_2 - 2y_1 + 1y_0 \geq 5 \\
6 & -1x_3 + 2x_2 + 1x_1 - 2x_0 - 2y_3 - 1y_2 + 2y_1 + 1y_0 \geq 4 \\
7 & +1x_3 - 2x_2 + 3x_1 - 3x_0 - 3y_3 - 3y_2 - 1y_1 - 2y_0 \geq 11 \\
8 & +3x_3 - 1x_2 - 1x_1 - 3x_0 - 2y_3 + 1y_2 + 2y_1 + 2y_0 \geq 4 \\
9 & +2x_3 + 2x_2 - 1x_1 - 2x_0 - 2y_3 - 1y_2 - 1y_1 - 1y_0 \geq 6 \\
10 & +4x_3 + 2x_2 + 2x_1 + 4x_0 - 1y_3 - 1y_2 + 2y_1 - 2y_0 \geq 0 \\
11 & +2x_3 + 2x_2 - 2x_1 + 3x_0 - 3y_3 - 1y_2 - 4y_1 + 4y_0 \geq 6 \\
12 & +1x_3 - 1x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 - 2y_1 - 1y_0 \geq 3 \\
13 & +1x_3 - 3x_2 - 1x_1 - 2x_0 + 2y_3 + 3y_2 + 0y_1 + 3y_0 \geq 3 \\
14 & -2x_3 - 2x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 + 2y_1 + 2y_0 \geq 2 \\
15 & -2x_3 - 2x_2 + 2x_1 + 1x_0 - 1y_3 + 1y_2 - 1y_1 + 2y_0 \geq 4 \\
16 & -1x_3 - 2x_2 - 1x_1 - 2x_0 - 2y_3 + 1y_2 - 1y_1 - 2y_0 \geq 9 \\
17 & -2x_3 + 0x_2 - 2x_1 + 1x_0 - 1y_3 + 1y_2 + 2y_1 - 2y_0 \geq 5 \\
18 & -1x_3 + 2x_2 - 3x_1 - 1x_0 - 1y_3 - 3y_2 - 2y_1 + 3y_0 \geq 8
\end{array}$$

Secondary structure following Sun et al.'s [34] method (5 auxiliary constraints):

$$\begin{array}{ll}
1 & x_0 + x_1 + x_2 + x_3 - S \geq 0 \\
2 & \\
3 & x_0 - S \leq 0 \\
4 & x_1 - S \leq 0 \\
5 & x_2 - S \leq 0 \\
6 & x_3 - S \leq 0 \\
7 & \\
8 & -2x_3 + 1x_2 + 0x_1 - 2x_0 + 2y_3 + 2y_2 + 1y_1 - 1y_0 \geq 3 \\
9 & +2x_3 + 0x_2 + 1x_1 - 1x_0 + 2y_3 - 2y_2 + 2y_1 - 2y_0 \geq 3 \\
10 & +1x_3 + 3x_2 + 2x_1 - 3x_0 - 2y_3 + 2y_2 + 2y_1 + 3y_0 \geq 1 \\
11 & +1x_3 + 3x_2 + 3x_1 + 2x_0 - 1y_3 + 0y_2 - 2y_1 - 2y_0 \geq 2 \\
12 & -1x_3 + 2x_2 + 0x_1 - 2x_0 + 2y_3 - 2y_2 - 2y_1 + 1y_0 \geq 5 \\
13 & -1x_3 + 2x_2 + 1x_1 - 2x_0 - 2y_3 - 1y_2 + 2y_1 + 1y_0 \geq 4 \\
14 & +1x_3 - 2x_2 + 3x_1 - 3x_0 - 3y_3 - 3y_2 - 1y_1 - 2y_0 \geq 11 \\
15 & +3x_3 - 1x_2 - 1x_1 - 3x_0 - 2y_3 + 1y_2 + 2y_1 + 2y_0 \geq 4 \\
16 & +2x_3 + 2x_2 - 1x_1 - 2x_0 - 2y_3 - 1y_2 - 1y_1 - 1y_0 \geq 6 \\
17 & +4x_3 + 2x_2 + 2x_1 + 4x_0 - 1y_3 - 1y_2 + 2y_1 - 2y_0 \geq 0 \\
18 & +2x_3 + 2x_2 - 2x_1 + 3x_0 - 3y_3 - 1y_2 - 4y_1 + 4y_0 \geq 6 \\
19 & +1x_3 - 1x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 - 2y_1 - 1y_0 \geq 3 \\
20 & +1x_3 - 3x_2 - 1x_1 - 2x_0 + 2y_3 + 3y_2 + 0y_1 + 3y_0 \geq 3 \\
21 & -2x_3 - 2x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 + 2y_1 + 2y_0 \geq 2 \\
22 & -2x_3 - 2x_2 + 2x_1 + 1x_0 - 1y_3 + 1y_2 - 1y_1 + 2y_0 \geq 4 \\
23 & -1x_3 - 2x_2 - 1x_1 - 2x_0 - 2y_3 + 1y_2 - 1y_1 - 2y_0 \geq 9 \\
24 & -2x_3 + 0x_2 - 2x_1 + 1x_0 - 1y_3 + 1y_2 + 2y_1 - 2y_0 \geq 5 \\
25 & -1x_3 + 2x_2 - 3x_1 - 1x_0 - 1y_3 - 3y_2 - 2y_1 + 3y_0 \geq 8
\end{array}$$

Secondary structure following Bakshi's [4] method (1 auxiliary constraint):

$$\begin{array}{ll}
1 & x_0 + x_1 + x_2 + x_3 + y_0 + y_1 + y_2 + y_3 - S \geq 0 \\
2 & \\
3 & -2x_3 + 1x_2 + 0x_1 - 2x_0 + 2y_3 + 2y_2 + 1y_1 - 1y_0 + 3(1 - S) \geq 3 \\
4 & +2x_3 + 0x_2 + 1x_1 - 1x_0 + 2y_3 - 2y_2 + 2y_1 - 2y_0 + 3(1 - S) \geq 3 \\
5 & +1x_3 + 3x_2 + 2x_1 - 3x_0 - 2y_3 + 2y_2 + 2y_1 + 3y_0 + 1(1 - S) \geq 1 \\
6 & +1x_3 + 3x_2 + 3x_1 + 2x_0 - 1y_3 + 0y_2 - 2y_1 - 2y_0 + 2(1 - S) \geq 2 \\
7 & -1x_3 + 2x_2 + 0x_1 - 2x_0 + 2y_3 - 2y_2 - 2y_1 + 1y_0 + 5(1 - S) \geq 5 \\
8 & -1x_3 + 2x_2 + 1x_1 - 2x_0 - 2y_3 - 1y_2 + 2y_1 + 1y_0 + 4(1 - S) \geq 4 \\
9 & +1x_3 - 2x_2 + 3x_1 - 3x_0 - 3y_3 - 3y_2 - 1y_1 - 2y_0 + 11(1 - S) \geq 11 \\
10 & +3x_3 - 1x_2 - 1x_1 - 3x_0 - 2y_3 + 1y_2 + 2y_1 + 2y_0 + 4(1 - S) \geq 4 \\
11 & +2x_3 + 2x_2 - 1x_1 - 2x_0 - 2y_3 - 1y_2 - 1y_1 - 1y_0 + 6(1 - S) \geq 6 \\
12 & +4x_3 + 2x_2 + 2x_1 + 4x_0 - 1y_3 - 1y_2 + 2y_1 - 2y_0 + 0(1 - S) \geq 0 \\
13 & +2x_3 + 2x_2 - 2x_1 + 3x_0 - 3y_3 - 1y_2 - 4y_1 + 4y_0 + 6(1 - S) \geq 6 \\
14 & +1x_3 - 1x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 - 2y_1 - 1y_0 + 3(1 - S) \geq 3 \\
15 & +1x_3 - 3x_2 - 1x_1 - 2x_0 + 2y_3 + 3y_2 + 0y_1 + 3y_0 + 3(1 - S) \geq 3 \\
16 & -2x_3 - 2x_2 + 0x_1 + 1x_0 + 1y_3 + 0y_2 + 2y_1 + 2y_0 + 2(1 - S) \geq 2 \\
17 & -2x_3 - 2x_2 + 2x_1 + 1x_0 - 1y_3 + 1y_2 - 1y_1 + 2y_0 + 4(1 - S) \geq 4 \\
18 & -1x_3 - 2x_2 - 1x_1 - 2x_0 - 2y_3 + 1y_2 - 1y_1 - 2y_0 + 9(1 - S) \geq 9 \\
19 & -2x_3 + 0x_2 - 2x_1 + 1x_0 - 1y_3 + 1y_2 + 2y_1 - 2y_0 + 5(1 - S) \geq 5 \\
20 & -1x_3 + 2x_2 - 3x_1 - 1x_0 - 1y_3 - 3y_2 - 2y_1 + 3y_0 + 8(1 - S) \geq 8
\end{array}$$

The following observations can be made regarding the two methods. Bakshi's method [4] takes less number of auxiliary constraints. It requires only 1 auxiliary constraint, even though it modifies all the original constraints by inserting an additional term involving S . Sun et al.'s method [34], in contrast, requires more auxiliary constraints ($n + 1$ for an n -bit SBox).

Further, both methods [4, 34] inherently deal with 0th row and 0th column issues. For instances, in Sun et al.'s method; the first auxiliary constraint ensures that when all x_i are zero, S must be set to 0 (inactive). The next n auxiliary constraints ensure that if any x_i is non-zero, S can be set to 1 (active); and for bijective SBoxes the same constraints also ensure that at least one of the y_i is non-zero. $\square$

Example 8 (Negative sample handling in secondary structure). S is a binary variable which indicates if the SBox is active (with respect to negative samples), following the secondary structure modeling from [34] (the alternate method from [4] is omitted here for brevity). Similar to Example 7, the objective function remains the same, to maximize S .

$$\begin{array}{ll}
1 & x_0 + x_1 + x_2 + x_3 - (1 - S) \geq 0 \\
2 & \\
3 & x_0 - (1 - S) \leq 0 \\
4 & x_1 - (1 - S) \leq 0 \\
5 & x_2 - (1 - S) \leq 0 \\
6 & x_3 - (1 - S) \leq 0 \\
7 & \\
8 & -2x_3 + 1x_2 + 0x_1 - 2x_0 + 2y_3 + 2y_2 + 1y_1 - 1y_0 \geq 3 \\
9 & +2x_3 + 0x_2 + 1x_1 - 1x_0 + 2y_3 - 2y_2 + 2y_1 - 2y_0 \geq 3 \\
10 & +1x_3 + 3x_2 + 2x_1 - 3x_0 - 2y_3 + 2y_2 + 2y_1 + 3y_0 \geq 1 \\
11 & +1x_3 + 3x_2 + 3x_1 + 2x_0 - 1y_3 + 0y_2 - 2y_1 - 2y_0 \geq 2 \\
12 & -1x_3 + 2x_2 + 0x_1 - 2x_0 + 2y_3 - 2y_2 - 2y_1 + 1y_0 \geq 5
\end{array}$$

```

13 -1x3 +2x2 +1x1 -2x0 -2y3 -1y2 +2y1 +1y0 >= 4
14 +1x3 -2x2 +3x1 -3x0 -3y3 -3y2 -1y1 -2y0 >= 11
15 +3x3 -1x2 -1x1 -3x0 -2y3 +1y2 +2y1 +2y0 >= 4
16 +2x3 +2x2 -1x1 -2x0 -2y3 -1y2 -1y1 -1y0 >= 6
17 +4x3 +2x2 +2x1 +4x0 -1y3 -1y2 +2y1 -2y0 >= 0
18 +2x3 +2x2 -2x1 +3x0 -3y3 -1y2 -4y1 +4y0 >= 6
19 +1x3 -1x2 +0x1 +1x0 +1y3 +0y2 -2y1 -1y0 >= 3
20 +1x3 -3x2 -1x1 -2x0 +2y3 +3y2 +0y1 +3y0 >= 3
21 -2x3 -2x2 +0x1 +1x0 +1y3 +0y2 +2y1 +2y0 >= 2
22 -2x3 -2x2 +2x1 +1x0 -1y3 +1y2 -1y1 +2y0 >= 4
23 -1x3 -2x2 -1x1 -2x0 -2y3 +1y2 -1y1 -2y0 >= 9
24 -2x3 +0x2 -2x1 +1x0 -1y3 +1y2 +2y1 -2y0 >= 5
25 -1x3 +2x2 -3x1 -1x0 -1y3 -3y2 -2y1 +3y0 >= 8

```

However, an alternate version of this model can be constructed by simply running for the minimum number of active negative samples instead of maximum number of active positive samples. In that case, no auxiliary constraints are needed, and the objective function in that case would be to minimize S . Finally, to revert back to the original maximization problem, one has to subtract that from the total number of SBoxes. $\square$

B Reference to SBoxes

The following SBoxes are used in our experiments:

- | | | | |
|----------------|------------------|------------------|---------------|
| – ASCON [20] | – KLEIN [21] | – PICCOLO [31] | – SKINNY [9] |
| – FIDES-5 [13] | – LILLIPUT [10] | – PRESENT [15] | – TWINE [35] |
| – GIFT [8] | – MIBS [22] | – PRIDE [3] | – SC2000 [32] |
| – ICEPOLE [26] | – MIDORI [7] | – PRINCE [16] | |
| – KECCAK [11] | – MINALPHER [19] | – RECTANGLE [37] | |

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